

**Smart Learn: An Adaptive Mobile Platform for
Neurodevelopmental Learning Disorders**

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Project Proposal Report

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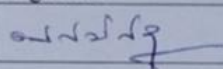
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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Working memory is an essential cognitive function that enables children to temporarily store, process, and use information during learning activities such as following instructions, remembering sequences, solving simple problems, and completing classroom tasks. For children aged 6–8 years, weaknesses in working memory can negatively affect academic performance, attention, confidence, and day-to-day learning. Although early support is important, there is a lack of accessible and localized digital systems designed specifically for Sinhala-speaking children with working memory difficulties.

This research proposes a Sinhala web-based mobile application within the Smart_Learn+ platform to support and improve working memory skills in children aged 6–8 years who have already been identified as having working memory difficulties. The system includes multiple structured working memory activities such as sequence recall, pattern matching, object recall, card matching, puzzle activities, and other mini-games. The child begins with basic-level tasks, and if the child becomes stuck during any activity, the system reduces the difficulty and provides simpler tasks to help the child continue successfully.

The novelty of this research is the AI-Assisted Adaptive Working Memory Game Generation Mechanism, which is specifically applied to the puzzle-game component. This mechanism analyzes the child's response time, error count, retries, task success, and performance trends in order to automatically generate or configure suitable puzzle-based activities according to the child's current ability level. In addition, the platform also includes progress monitoring dashboards that allow parents and therapists to track performance trends and support the child's learning process both at home and in educational settings.

The system is proposed to be developed using React.js for the user interface, Node.js with Express.js for backend services, and MongoDB or Firebase for secure data storage. The expected outcome of this research is an intelligent and accessible cognitive training platform that enhances working memory performance while increasing engagement through game-based learning. Socially, the system promotes inclusive and accessible cognitive support for Sinhala-speaking children, while commercially it has potential as a scalable and affordable digital intervention tool for schools, parents, and therapy centers.

Keywords: Working Memory, Adaptive Learning, Educational Technology, Sinhala Learning Support, AI-Based Learning System

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LIST OF ABBREVIATIONS

Abbreviation	Word / Description
NLDs	Neurodevelopmental Learning Disorders
WM	Working Memory
SDGs	Sustainable Development Goals
DSR	Design Science Research
AI	Artificial Intelligence
UI	User Interface
DB	Database
API	Application Programming Interface
VS Code	Visual Studio Code
IDE	Integrated Development Environment

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1. INTRODUCTION

1.1 Background and Context

Working memory is a core cognitive function that enables individuals to temporarily store and manipulate information while performing tasks such as learning, reasoning, and comprehension [1]. For children aged 6–8 years, working memory is especially important for early academic development because it supports skills such as following multi-step instructions, remembering sequences, solving basic mathematical problems, and understanding reading content [1], [2]. Children with working memory difficulties often struggle to retain and process information, which can lead to forgotten instructions, repeated errors, and incomplete classroom activities even when their general intelligence is within the normal range [2], [4].

These difficulties are particularly significant during the early primary-school years, which represent a critical period for cognitive and educational development. Children aged 6–8 are developing foundational abilities in reading comprehension, number sequencing, and problem-solving. Weak working memory can cause children to forget instructions, struggle with multi-step tasks, and fail to retain information during learning activities. Such challenges can negatively affect academic performance, self-confidence, and motivation, and may lead to long-term educational difficulties if early support is not provided [2], [4].

Globally, research indicates that around 10–15% of children in regular classrooms experience working memory difficulties that negatively affect academic learning and progress [3]. Children with low working memory are also more likely to experience difficulties in areas such as reading and mathematics compared to peers with stronger memory skills [4]. For Sinhala-speaking contexts such as Sri Lanka, this challenge is made more difficult by the limited availability of localized digital tools, as many widely discussed working memory and brain-training systems are designed primarily in English [10], [15], [18].

Existing interventions include therapist-guided training, structured exercises, and digital applications with memory games, sequence recall tasks, and attention-based activities [5]. However, most existing applications are designed in English, which limits accessibility for Sinhala-speaking children. In addition, many systems require continuous supervision, focus on broad cognitive training rather than child-specific support, and do not provide sufficient personalization and caregiver monitoring within a single platform [10], [15], [18].

To address these limitations, this research proposes a Sinhala web-based mobile application to support children aged 6–8 who have already been identified as having working memory difficulties. The system provides several structured working memory activities such as sequence recall, pattern matching, object recall, card matching, puzzle activities, and other mini-games. These activities can be used under therapist supervision or independently with parental guidance, making the platform suitable for both guided intervention and home-based practice.

The overall system begins with basic-level activities. If the child becomes stuck during any activity, the system automatically reduces the difficulty level and provides simpler tasks to help

the child continue learning gradually and successfully. This supportive adaptation mechanism is designed to reduce frustration, maintain engagement, and encourage continued participation in the learning process.

The novelty of this research lies in the AI-Assisted Adaptive Working Memory Game Generation Mechanism, which is specifically applied to the puzzle activity component. This mechanism dynamically generates or configures puzzle-based tasks according to the child's current memory weaknesses, performance patterns, response time, errors, and task behavior. By adapting puzzle activities in this way, the system provides more personalized support while the other working memory activities remain available as structured training tasks within the platform.

The platform also provides progress dashboards for therapists and parents, enabling them to monitor performance trends, identify learning difficulties, observe level progression, and guide the child's learning both during therapy sessions and at home. In this way, the system combines structured cognitive activities, supportive task adaptation, performance tracking, and caregiver monitoring within a single platform.

This research supports several SDGs, including SDG 4: Quality Education by offering inclusive and accessible learning tools, SDG 3: Good Health and Well-being by promoting early cognitive development, and SDG 10: Reduced Inequalities by providing an affordable digital platform accessible to children from various socio-economic backgrounds [6].

The proposed system also aligns with the FoC Research Cluster: Technology Integration & Management, as it combines educational technology, rule-based artificial intelligence, web-based system development, and digital monitoring tools to create a scalable and accessible intervention platform. By integrating adaptive support mechanisms with culturally relevant Sinhala-language content, the system addresses the research gap in digital cognitive training tools designed specifically for Sinhala-speaking children with working memory difficulties.

1.2 Literature Review

Working memory (WM) is a fundamental cognitive system that enables individuals to temporarily store and manipulate information needed for learning, reasoning, comprehension, and task completion [1]. In children aged 6–8 years, working memory plays an important role in remembering instructions, following sequences, solving simple mathematical tasks, and understanding reading content [1], [2]. Studies have shown that children with poor working memory often experience difficulties in reading, mathematics, classroom attention, and task completion even when their overall intelligence is within the normal range [2], [4]. These findings show the importance of early identification and intervention for children with working memory difficulties.

Traditional approaches to working memory support commonly include therapist-led exercises such as digit span practice, sequence recall, repetition drills, and structured memory activities

[5]. Although such methods can be useful, they are often time-consuming, require continuous adult supervision, and are difficult to scale across home and school environments. As a result, digital and game-based interventions have increasingly been explored as alternative methods for improving working memory in children. Gamified learning environments are especially valuable because they can increase motivation, engagement, and persistence during repeated cognitive practice [9].

Several digital systems and applications have addressed working memory training or broader cognitive development. Cogmed is a structured working memory training program that can be tailored for users from age 4 and above, but it is mainly designed as a commercial intervention protocol rather than a localized child-centered learning platform [15]. Lumosity provides adaptive brain-training games and performance insights, but it is not designed for children under 13 years of age and does not specifically focus on working memory intervention for young children [16], [18]. Mind Games offers a collection of general brain-training activities together with score history and progress graphs, but it mainly supports broad self-training rather than structured intervention for young children with working memory difficulties [17]. In addition, WorM is best understood as a research-based serious game prototype developed to assess and train working memory in children with neurodevelopmental disorders, rather than a fully deployed commercial or language-localized platform [10].

Despite these developments, several important limitations remain. Many existing systems are designed primarily in English and focus on general cognitive training or fixed task progression rather than providing culturally relevant support for young children in local educational contexts. Furthermore, many tools provide only basic progress summaries and do not offer detailed behavioral insights for parents, caregivers, or therapists, such as response time trends, retry behavior, error patterns, or child-specific performance changes over time [10], [15], [18]. Research also suggests that the transfer of working memory training to broader academic outcomes is often mixed, indicating the need for more personalized, age-appropriate, and engaging interventions [12].

Another limitation is that existing systems usually provide static games, predefined difficulty levels, or broad adaptive training, but they do not clearly support a mechanism that generates or configures puzzle-based working memory tasks according to a child's real-time performance behavior in a localized child-centered environment. While some systems include adaptive features, these are not typically presented as a targeted puzzle-generation mechanism for young Sinhala-speaking children with caregiver monitoring. Based on the systems reviewed in this proposal, the combination of localized delivery, caregiver dashboards, and AI-assisted adaptive puzzle generation is not clearly available in the compared platforms [10], [15], [18].

To address these limitations, the proposed system introduces a Sinhala web-based mobile platform for children aged 6–8 years with working memory difficulties. The system includes multiple working memory activities such as sequence recall, pattern matching, object recall, card matching, and other mini-games that support regular cognitive practice. However, the core novelty of the proposed system lies specifically in the AI-Assisted Adaptive Working Memory Game Generation Mechanism, which is applied to the puzzle-game component. This

mechanism analyzes the child’s gameplay data, including response time, error count, retries, task success, and behavioral trends, and then generates or configures suitable puzzle-based activities according to the child’s current working memory performance.

This distinction is important because the proposed system does not claim that all included activities are AI-generated. Instead, the broader platform provides a range of structured working memory activities, while the novelty is focused on adaptive puzzle generation as a personalized intervention strategy. In addition, the system integrates Sinhala instructions, supportive difficulty reduction when a child becomes stuck, and dashboard-based monitoring for parents and therapists. These features make the intervention more accessible, understandable, and practical for both independent home use and guided therapeutic support.

In summary, existing literature and current digital applications demonstrate that game-based approaches can support working memory development, but important gaps remain in localization, caregiver monitoring, and highly personalized adaptive activity design. The proposed system addresses these gaps by combining multiple working memory activities with a novel AI-Assisted Adaptive Working Memory Game Generation Mechanism for puzzle-based tasks, offering a more child-centered and culturally relevant intervention model for Sinhala-speaking children aged 6–8 years.

1.3 Research Gap

Although several digital tools and research studies have explored working memory training, important gaps still remain in the design of accessible and personalized support systems for young children, particularly for Sinhala-speaking learners. Existing platforms such as Cogmed [15], Lumosity [16], [18], Mind Games [17], and research-based systems like WorM [10] demonstrate that game-based working memory activities can support engagement and practice. However, these systems do not adequately address the combined needs of language accessibility, child-specific personalization, caregiver monitoring, and adaptive intervention within a single integrated platform.

A major limitation is that most current systems are designed primarily in English, which reduces accessibility and usability for Sinhala-speaking children in Sri Lanka. Language barriers can reduce comprehension of instructions, lower engagement, and make independent use more difficult for children in the early primary-school age range. Furthermore, many existing systems target broad cognitive training rather than the specific support needs of children aged 6–8 with working memory difficulties [10], [15], [18].

Another important gap lies in personalization. Many existing systems rely on fixed tasks, predefined levels, or limited adaptation. Even when adaptive elements are present, they are not usually designed to generate suitable next-step puzzle activities according to a child’s detailed performance behavior. Therefore, a more responsive mechanism is needed to analyze task

accuracy, response time, retry behavior, and error patterns in order to generate puzzle activities that match the child’s current ability level [10], [15], [18].

In addition, many existing tools provide only simple performance summaries and do not offer detailed progress tracking for parents, caregivers, or therapists. As a result, adults supporting the child may not receive enough information to understand improvement trends, identify persistent weaknesses, or guide intervention decisions effectively [10], [15], [18].

The novelty of this research lies in the AI-Assisted Adaptive Working Memory Game Generation Mechanism, which is specifically applied to the puzzle-game component. This mechanism automatically generates or configures puzzle-based working memory activities according to the child’s current performance trend, errors, response time, and task behavior. At the same time, the overall platform includes other structured working memory activities such as sequence recall, pattern matching, object recall, card matching, and other mini-games, together with supportive difficulty reduction when the child becomes stuck and dashboard-based monitoring for parents and therapists.

Therefore, the core research gap is the absence of a Sinhala-language, child-centered, and fully monitored working memory support platform that can generate adaptive puzzle-based game activities for children aged 6–8 based on real-time performance behavior, while also providing a broader set of structured working memory training activities and caregiver monitoring features.

Table 1: System Comparison

Feature / System	Cogmed	Lumosity	Mind Games	WorM (Research)	Smart_Learn+
Age Suitability (6–8)	<i>Limited</i>	✗	✗	<i>Limited</i>	✓
Sinhala Support	✗	✗	✗	✗	✓
Adaptive Learning	✓	✓	<i>Limited</i>	<i>Limited</i>	✓
Working Memory Activities	✓	<i>Limited</i>	<i>Limited</i>	✓	✓
Variety of Games	<i>Limited</i>	✓	✓	<i>Limited</i>	✓
AI Puzzle Generation (novelty)	✗	✗	✗	✗	✓
Voice Instructions	✗	✗	✗	✗	✓

Progress Tracking	✓	✓	✓	Limited	✓
Parent Dashboard	Limited	✗	✗	Limited	✓
Personalization	Limited	Limited	✗	Limited	✓

2. OBJECTIVE

2.1 Main Objective

The main objective of this research is to develop and evaluate an AI-driven adaptive Sinhala web-based mobile application within the Smart_Learn+ platform that supports children aged 6–8 years with working memory difficulties by providing structured working memory activities, supportive difficulty reduction when a child becomes stuck, and an AI-Assisted Adaptive Working Memory Game Generation Mechanism for puzzle-based activities, together with performance dashboards for parents and therapists during the defined intervention period.

This objective follows the SMART criteria:

- **Specific:** Focuses on working memory support using Sinhala-language, game-based activities, supportive task adaptation, and AI-assisted puzzle generation.
- **Measurable:** Performance improvement will be measured using task accuracy, response time, error frequency, retry count, completion rate, and dashboard-based progress tracking.
- **Achievable:** Implemented using web-based mobile technologies, backend adaptive logic, secure data storage, and structured activity templates.
- **Relevant:** Directly addresses working memory difficulties while providing culturally and linguistically appropriate support for Sinhala-speaking children.
- **Time-bound:** Developed and evaluated within the project implementation period

2.2 Specific Objectives

The specific objectives of this research are:

- To design and develop a Sinhala web-based mobile application that delivers interactive memory games suitable for children aged 6–8, ensuring usability, engagement, and cultural relevance.
- To implement structured working memory activities such as sequence recall, pattern matching, object recall, card matching, puzzle activities, and other mini-games.
- To develop the AI-Assisted Adaptive Working Memory Game Generation Mechanism for the puzzle-game component, using response time, error count, retries, and task behavior to generate or configure suitable puzzle-based activities.
- To create a performance analytics and progress-monitoring module that records activity data and provides dashboards for parents and therapists.

- To develop personalized memory training activities tailored to the child’s cognitive profile, progressively increasing in complexity to enhance working memory capacity over the intervention period.
- To provide monitoring and feedback dashboards for parents and therapists, enabling them to track the child’s progress, identify difficulties, and guide interventions both at home and in therapy sessions.
- To evaluate the effectiveness of the system by measuring improvements in working memory performance (accuracy, response time, task completion rate) before, during, and after the 12-week intervention period.

3. Methodology

3.1 Research Approach

This research follows a Design Science Research (DSR) approach, which is suitable for developing innovative, artifact-based solutions to real-world problems. The study aims to design, implement, and evaluate an AI-driven adaptive Sinhala web-based mobile application for children aged 6–8 with working memory difficulties. The DSR approach allows iterative design, testing, and refinement of the system while integrating theoretical knowledge about working memory and adaptive learning principles.

An experimental evaluation is embedded within the DSR framework to assess the effectiveness of the intervention. Children’s working memory performance will be measured before, during, and after the intervention to quantify the impact of the system.

3.2 Technical Methods and Algorithms

The proposed system uses several technical methods to support working memory training and progress monitoring.

Performance Data Collection

The system will collect the child’s gameplay data while completing activities. These data include:

- accuracy
- response time
- error count
- retry count
- task completion status

These data help the system understand how well the child is performing.

Weighted Performance Analysis

A weighted scoring method will be used to evaluate the child's performance. This score will be calculated using factors such as accuracy, time taken, number of errors, and retries. Based on this score, the system can identify whether the child is doing well, facing difficulties, or needing easier support.

Difficulty Adjustment Mechanism

The system will include a difficulty adjustment mechanism for all activities. At the beginning, the child will receive easy tasks. If the child performs well, the system will gradually increase the difficulty. If the child becomes stuck or makes repeated mistakes, the system will reduce the difficulty and give a simpler task. This helps the child continue without frustration.

AI-Assisted Adaptive Working Memory Game Generation Mechanism

The main novelty of the system is the AI-Assisted Adaptive Working Memory Game Generation Mechanism, which is applied only to the puzzle activity. This mechanism will use the child's response time, errors, retries, task success, and recent performance trend to generate or configure a suitable puzzle-based activity.

The system will not create all games from scratch. Instead, it will use a set of puzzle templates and adjust them by changing:

- number of items
- hint level
- time allowance
- similarity of objects
- distractor count
- difficulty level

This helps provide puzzle tasks that match the child's current working memory ability.

Structured Working Memory Activities

In addition to the adaptive puzzle activity, the system will also provide other structured activities such as:

- sequence recall
- pattern matching
- object recall
- card matching
- other mini-games

These activities are included to support regular working memory practice.

Progress Monitoring

The system will store the child’s activity data and performance history in a database. These data will be shown in parent and therapist dashboards. The dashboards will display:

- accuracy
- response time
- completed activities
- current level
- progress over time

This helps adults track the child’s development more easily.

3.3 Tools and Platforms

Table 2: tools and platforms

Component	Tool/Platform	Justification
Web-based mobile app	ReactJS	Supports responsive, cross-platform applications compatible with Android and iOS browsers.
Backend & database	Node.js, Express, MongoDB	Provides scalable storage of user performance data and supports real-time analytics.
AI/adaptive logic	Python with rule-based logic	Facilitates dynamic task adaptation using algorithms for weighted scoring and performance classification.
Dashboards	Chart.js / Recharts in ReactJS	Visualizes performance metrics for therapists and parents in an interactive manner.
Database	MongoDB / Firebase	Stores student profiles, pronunciation results, response times, and learning progress data securely and efficiently.

For development and collaboration, Git is used for version control, with the project repository hosted on GitHub. The system is developed using Visual Studio Code (VS Code), which serves as the main development environment for coding, debugging, and project management.

3.4 Data Collection and Ethical Considerations

The data collection process focuses on gathering expert knowledge from educational and cognitive development professionals and collecting performance data related to working memory abilities, rather than obtaining sensitive personal information from children. This approach minimizes ethical concerns while maintaining research validity. The data collected

includes key working memory performance indicators such as accuracy in recall tasks, response time, error patterns, and sequence retention during memory-based activities. These indicators help evaluate the child's working memory capacity and attention control.

In addition, expert insights were gathered from educational psychologists, teachers, and learning support specialists to identify common working memory difficulties experienced by children aged 6–7 years. These difficulties include challenges in following multi-step instructions, remembering sequences, maintaining attention during tasks, and recalling recently presented information.

Based on this expert knowledge, structured caregiver questionnaires, working memory assessment games, and adaptive training activities were designed. The collected expert input also helped define severity levels of working memory difficulties categorized as mild, moderate, and severe. This information was used to guide the development of the performance analytics model and the weighted scoring algorithm used in the Smart_Learn+ system to evaluate children's working memory performance and provide personalized training recommendations

3.5 Validation Strategy and Performance Metrics

1. Validation Strategy:

- **Internal validation:** Testing AI adaptation algorithms and task progression logic to ensure correct difficulty adjustments.
- **Usability testing:** Conducted with children and caregivers to evaluate interface design, engagement, and ease of use.
- **Experimental evaluation:** Comparing pre- and post-intervention working memory scores to measure cognitive improvements.

2. Performance Metrics:

- **Accuracy Rate:** Percentage of correctly completed tasks.
- **Response Time:** Average time taken to complete each activity.
- **Error Frequency:** Number of incorrect attempts during activities.
- **Retry Count:** Number of repeated attempts required to complete tasks.
- **Task Completion Rate:** Percentage of started tasks that are completed successfully.
- **Progress Tracking:** Improvement trends visualized through parent and therapist dashboards.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

System Architecture

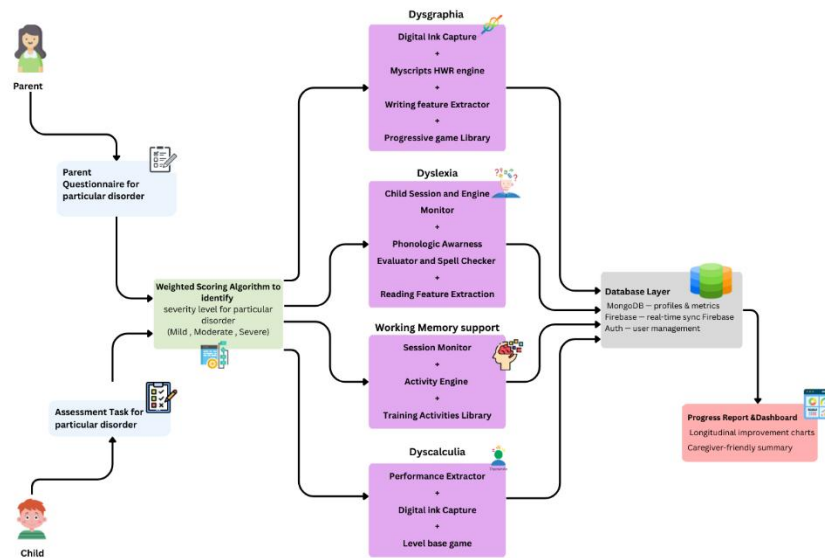


Figure 1: system diagram

Component Architecture

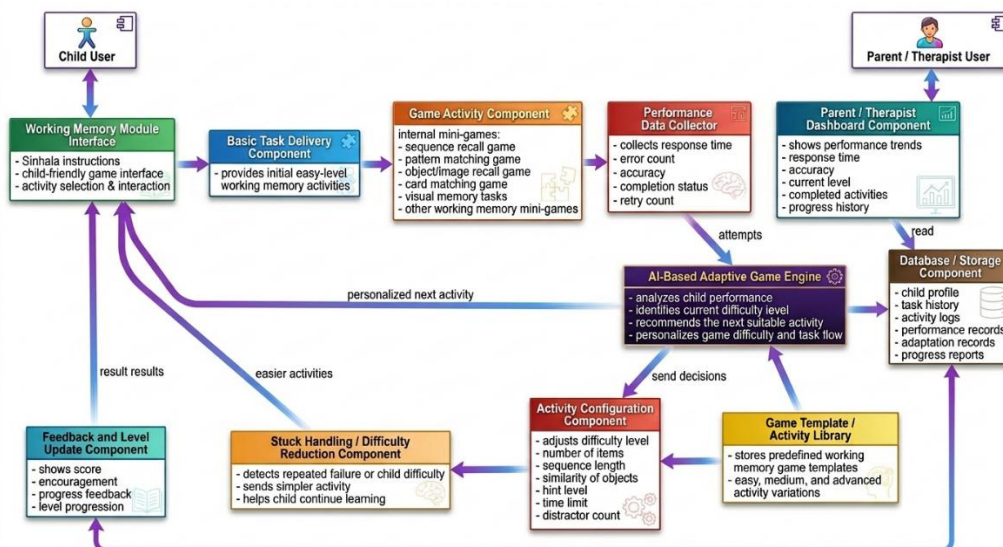


Figure 2: component diagram

5. USER REQUIREMENTS

5.1 Stakeholder Identification

Children (Primary Users)

Children aged 6–8 years with working memory difficulties will use the application to complete memory games and activities to improve memory skills.

Teachers / Therapists (Secondary Users)

Teachers and therapists will monitor children’s performance, identify difficulties, and guide memory training using the system’s dashboards.

Parents / Guardians (Tertiary Users)

Parents and guardians will supervise children at home, track progress through dashboards, and support learning activities.

5.2 Functional Requirements

1. Allow parents or guardians to register children and create individual profiles.
2. Allow the child to access the Working Memory support module after the disorder category has already been identified.
3. Provide structured working memory activities, including sequence recall, pattern matching, object recall, card matching, puzzle activities, and other mini-games.
4. Start the child with basic-level activities and record responses during each task.
5. Detect when the child is struggling or becomes stuck during an activity.
6. Automatically reduce difficulty and provide a simpler task when the child becomes stuck, so that the child can continue learning.
7. Analyze performance data such as accuracy, response time, retries, and error count.
8. Generate or configure personalized puzzle-based activities through the AI-Assisted Adaptive Working Memory Game Generation Mechanism.
9. Provide immediate feedback, encouragement, or hints when children make mistakes.
10. Store learning progress, task completion data, adaptation history, and performance analytics securely in a database.
11. Offer progress dashboards for parents and therapists to visualize trends, current level, and improvements over time.
12. Send notifications and reminders for scheduled sessions or incomplete tasks.
13. Support multi-device access, including mobile web browsers and desktop platforms

Non-Functional Requirements

Non-functional requirements describe how well the system performs and the quality attributes it must maintain.

1. Usability

The system shall provide a simple, colorful, and child-friendly user interface that enables children aged 6–8 years to easily interact with memory games and activities.

2. Performance

The system shall provide fast response times during memory game interactions and performance analysis to ensure smooth gameplay without noticeable delays.

3. Security

The system shall ensure that user data, including child profiles and performance records, is securely stored and protected from unauthorized access.

4. Reliability

The system shall operate consistently and reliably during training sessions, minimizing system crashes and preventing loss of progress data.

5. Accessibility

The system shall be accessible on commonly used smartphones, tablets, and web browsers, allowing children to use the application easily in both home and educational environments.

6. Scalability

The system architecture shall support future expansion, including the addition of new memory games, adaptive learning modules, and support for a larger number of users.

6. COMMERCIALIZATION PLAN

6.1 Target Market and Customer Persona

The primary target market for the proposed system includes children aged 6–8 years with working memory difficulties, therapists who support their cognitive development, and parents or guardians who guide their learning. The application is particularly valuable for Sinhala-medium educational environments, where language-specific intervention tools for working memory are limited.

The main customers who would adopt the system include:

1. Child User Persona

Field	Details
Persona Type	Child User
Name	Kavindu Fernando
Age	8 years
Background	Grade 3 student who likes mobile games and educational videos. He often forgets teacher instructions and has difficulty completing tasks with multiple steps.
Goals	Improve memory and attention during school activities; follow instructions more easily; enjoy learning through fun games.
Challenges	Difficulty remembering sequences and instructions; forgets homework steps; gets distracted easily.
How the System Helps	Provides memory games, sequence recall tasks, and pattern recognition activities. The AI adjusts difficulty based on performance to improve concentration and working memory.

1. Therapist Persona

Field	Details
Persona Type	Therapist
Name	Mrs. Chamari Wijesinghe
Age	34 years
Occupation	Child Therapist
Background	Mrs. Chamari works with young children who have learning and attention difficulties. She observes that some children struggle to remember instructions, follow sequences, and complete cognitive tasks independently.
Goals	Support children with working memory and attention difficulties; improve their ability to follow instructions and complete tasks; use engaging therapy tools for intervention.

Challenges	Limited time for one-to-one therapy; lack of digital cognitive training tools for young children; limited Sinhala-language intervention resources.
How the System Helps	The platform provides interactive memory games, adaptive activities, and progress reports that help therapists monitor each child's performance and identify areas needing additional support.

1. Parent Persona

Field	Details
Persona Type	Parent
Name	Dilshan Karunaratne
Age	38 years
Background	Father of a 7-year-old child who forgets homework instructions and classroom tasks. He wants to support his child but is unsure how to help effectively.
Goals	Improve the child's memory and learning ability; help the child complete schoolwork independently; use simple tools at home.
Challenges	Limited awareness of working memory problems; difficulty finding suitable apps; busy work schedule.
How the System Helps	Provides memory training games, adaptive difficulty, and progress dashboards so the child can practice independently while the parent monitors improvement.

6.2 Value Proposition

Table 3: value proposition

Value Element	Description
Affordable Cognitive Support	Reduces the need for frequent therapy sessions by providing daily practice at a fraction of the cost.
Clear Progress Visibility	Dashboards for parents and teachers show measurable improvements in memory tasks, response time, and task accuracy.
Inclusive Classroom Support	Enables teachers to support children with working memory difficulties without specialized training.

Time Savings	Reduces time spent on manual progress tracking and reporting for both parents and educators.
Engaging Game Experience	Memory exercises delivered as fun, interactive games with visual cues, animations, and rewards to maintain motivation.
Builds Confidence	Positive reinforcement and visible progress reduce frustration and improve self-esteem.
Age-Appropriate Design	Bright visuals, friendly characters, and Sinhala-language audio prompts designed for 6–8-year-olds.
Peace of Mind	Parents and educators can be confident that children are receiving targeted support for their cognitive development.

6.3 Revenue Model

The system will generate revenue through multiple models:

1. Subscription Model

- Monthly or yearly subscription for parents.
- Access to advanced analytics, personalized recommendations, and detailed reports.

2. Institutional Licensing

- Schools and therapy centers pay licensing fees.
- Access for multiple child profiles under one institutional account.

3. Freemium Model

- Basic memory games available for free.
- Advanced AI-based analytics and full reporting available under premium plans.

6.4 Cost Estimation

The estimated development and deployment costs include:

Table 4:cost

Cost Category	Estimated Cost (LKR)	Description
System Development	Rs. 50,000 - 70,000	Mobile app development, backend system, database integration
AI & Algorithm Development	Rs. 20,000 - 35,000	Performance analytics engine and weighted scoring implementation
UI/UX Design	Rs. 15,000 - 25,000	Child-friendly interface, animations, dashboard design
Cloud Hosting & Infrastructure	Rs. 10,000 - 20,000	Server hosting, database storage, domain setup
Expert Validation	Rs. 15,000 - 30,000	Consultation with educational experts and therapists
Testing & Pilot Study	Rs. 15,000 - 25,000	User testing, feedback collection, system improvement

Total Estimated Cost

Rs. 125,000 – 220,000

6.5 Pricing Strategy

A penetration pricing strategy will be used initially to attract early users and increase adoption.

Pricing Plans:

- **Basic Plan** – Low-cost monthly subscription with limited features.
- **Standard Plan** – Full intervention access with progress tracking and analytics.
- **Institutional Plan** – Discounted bulk pricing for schools and therapy centers with multiple user access.

Pricing will remain affordable to ensure accessibility in low-resource environments while maintaining long-term sustainability and revenue growth.

6.6 Competitive Advantage

The proposed system offers several competitive advantages compared to existing cognitive training applications.

Integrated Working Memory Support Platform

Combines structured working memory activities, adaptive support, analytics, and caregiver dashboards within a single system.

AI-Assisted Adaptive Working Memory Game Generation Mechanism

Introduces puzzle-based activity generation or configuration according to the child's response time, errors, retries, and performance trend.

Supportive Difficulty Reduction

When a child becomes stuck, the system provides simpler activities to maintain engagement and avoid frustration.

Sinhala Language Accessibility

Provides Sinhala instructions and voice guidance, making the platform more understandable and accessible for young Sinhala-speaking children.

Parent and Therapist Monitoring

Includes dashboards and progress reports that help adults monitor accuracy, response time, level progression, and ongoing weaknesses.

Specialized Focus on Children Aged 6–8 Years

Designed specifically for early primary-school children with working memory difficulties rather than for general adult brain training.

Many existing systems that provide either general cognitive games or limited static activities, the proposed system combines structured working memory tasks, supportive adaptation, localized language support, caregiver monitoring, and a novel AI-assisted puzzle-generation mechanism in a single child-centered platform.

6.7 Intellectual Property Considerations

Protecting intellectual property is important to ensure the long-term commercial value of the proposed system.

- **Software Copyright:** The system's source code, algorithms, and system architecture can be protected under software copyright.
- **Trademark Protection:** The application name and brand identity can be registered as a trademark to prevent unauthorized use.
- **Confidential Development Assets:** Core system logic, adaptive learning algorithms, and analytics methods should remain confidential.

- **Data Privacy Protection:** User data, especially children's learning records, must be securely stored and handled according to privacy and ethical standards.

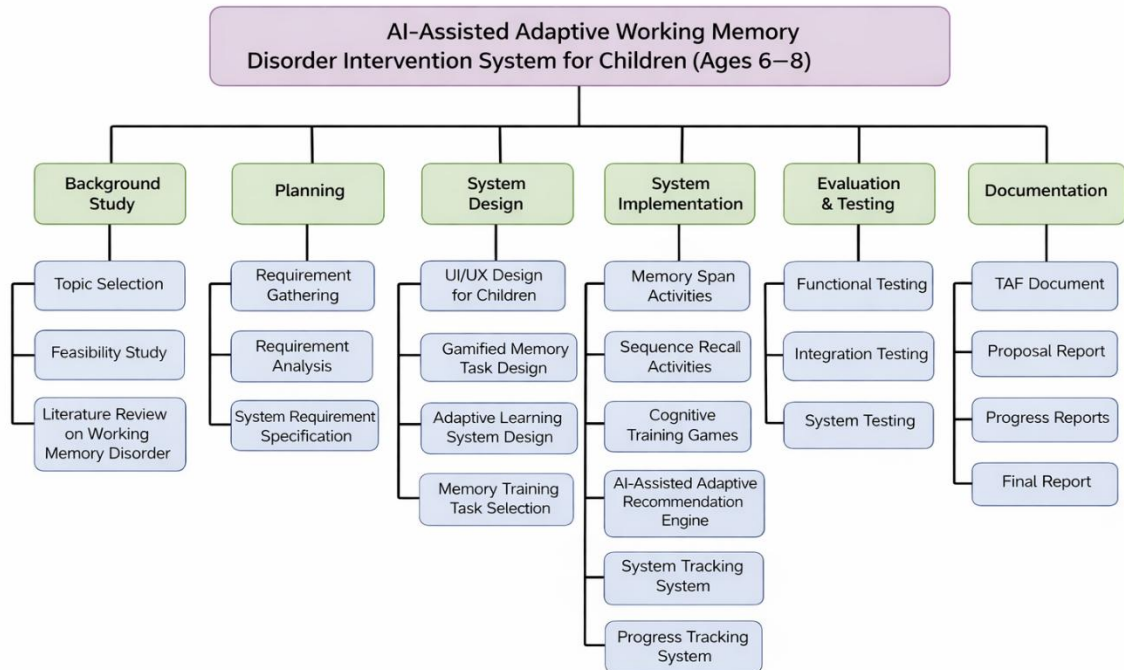
Protecting these intellectual assets increases the commercial value, credibility, and scalability of the system for future development and deployment.

7. BUDGET AND JUSTIFICATION

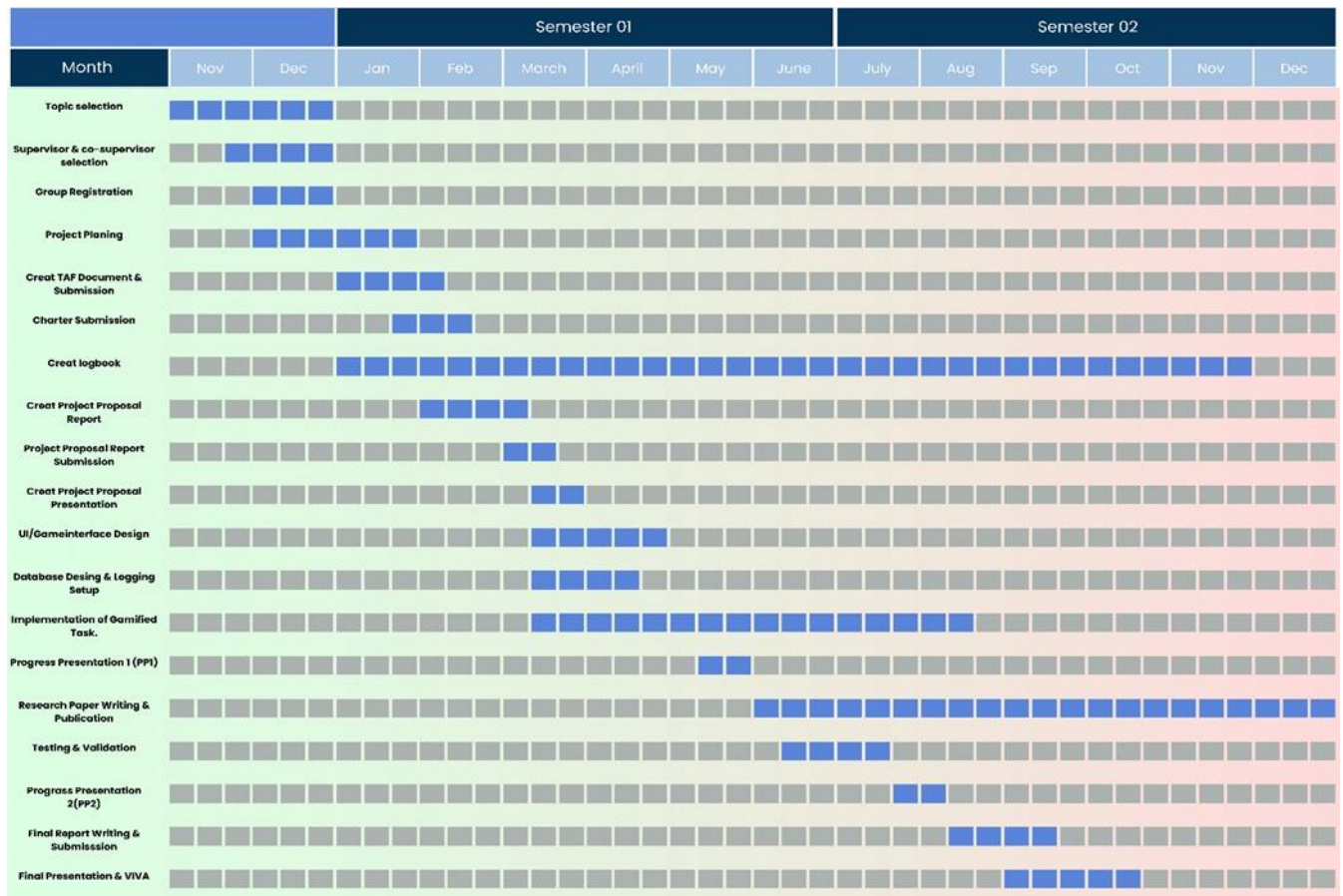
Table 5: budget

Cost Category	Estimated Cost (LKR)	Justification
Travel Expenses	9,000	Visits to SLIIT Matara Center for meetings with supervisor and co-supervisor. Visits to Ayati Center and other disability/therapy centers for expert consultation and validation.
Data Collection	10,000	Arranging appointments with experts and therapists. Printing assessment forms, questionnaires, and collecting feedback related to working memory evaluation.
Printing & Documentation	1,500	Printing proposal, progress reports, final report, consent forms, questionnaires, and other project documents.
Cloud Hosting / Server Costs	12,000	Hosting backend services, database storage, and system deployment during development and testing.
Internet & Communication	2,000	Internet usage for development, online meetings, system testing, and data transfer.

8. WORK BREAKDOWN STRUCTURE (WBS)



9. GANTT CHART



REFERENCES LIST

- [1] A. Baddeley, “Working memory: Theories, models, and controversies,” *Annual Review of Psychology*, vol. 63, pp. 1–29, 2012.
- [2] S. E. Gathercole and T. P. Alloway, *Working Memory and Learning: A Practical Guide for Teachers*. London, U.K.: Sage, 2008.
- [3] S. Colmar, J. D. Double, and S. D. F. Fraser, “Working memory interventions with children: Classrooms or computers?,” *Journal of Psychologists and Counsellors in Schools*, vol. 27, no. 2, pp. 264–276, 2017.
- [4] S. E. Gathercole, F. Woolgar, R. A. Kievit, D. Astle, T. Manly, and J. Holmes, “How common are working memory deficits in children with difficulties in reading and mathematics?,” *Journal of Applied Research in Memory and Cognition*, vol. 5, no. 4, pp. 384–394, 2016.
- [5] T. Klingberg, “Training and plasticity of working memory,” *Trends in Cognitive Sciences*, vol. 14, no. 7, pp. 317–324, 2010.
- [6] United Nations, “Goal 4: Quality Education,” *Sustainable Development Goals*. [Online]. Available: United Nations SDG website. [Accessed: Mar. 18, 2026].
- [7] United Nations, “Goal 3: Good Health and Well-being,” *Sustainable Development Goals*. [Online]. Available: United Nations SDG website. [Accessed: Mar. 18, 2026].
- [8] United Nations, “Goal 10: Reduced Inequalities,” *Sustainable Development Goals*. [Online]. Available: United Nations SDG website. [Accessed: Mar. 18, 2026].
- [9] W. J. Boendermaker, R. C. Veltkamp, and P. J. M. Prins, “Gamification of cognitive training: A comparison of working memory training and gamified working memory training in healthy adolescents,” *Journal of Cognitive Enhancement*, vol. 1, no. 4, pp. 419–427, 2017.
- [10] P. Ferreira, A. Amado, M. M. da Silva, T. S. Brito, and T. Dias, “Neurodevelopmental disorders: assessing and training working memory,” *Cognitive Research: Principles and Implications*, vol. 10, no. 1, 2025.
- [11] J. Holmes and S. E. Gathercole, “Taking working memory training from the laboratory into schools: The nature of individual differences in short-term training and classroom progress,” *Frontiers in Psychology*, vol. 5, 2014.
- [12] G. Sala and F. Gobet, “Cognitive training does not enhance general cognition,” *Trends in Cognitive Sciences*, vol. 23, no. 1, pp. 9–20, 2019.
- [13] T. P. Alloway and R. G. Alloway, *Understanding Working Memory: A Classroom Guide*, 2nd ed. London, U.K.: Sage, 2015.
- [14] K. P. Raghubar, M. A. Barnes, and S. A. Hecht, “Working memory and mathematics: A review of developmental, individual difference, and cognitive approaches,” *Learning and Individual Differences*, vol. 20, no. 2, pp. 110–122, 2010.

- [15] Cogmed, “Get Cogmed for your School / Special Education.” [Online]. Available: Cogmed official website. [Accessed: Mar. 18, 2026].
- [16] Lumosity Help Center, “Is Lumosity designed for children?” [Online]. Available: Lumosity Help Center. [Accessed: Mar. 18, 2026].
- [17] Mindware Consulting, “Mind Games,” *Google Play Store*. [Online]. Available: Google Play Store. [Accessed: Mar. 18, 2026].
- [18] Lumosity, “Lumosity: Brain Training Games / Personalized Training.” [Online]. Available: Lumosity official website and app listing. [Accessed: Mar. 18, 2026].

10. APPENDICES

Survey instruments

Smart Learn: An Adaptive Mobile Platform for Neurodevelopmental Learning Disorders Introduction

Smart Learn: An Adaptive Mobile Platform for Neurodevelopmental Learning Disorders is a research and development project conducted by final-year undergraduate students in Information Technology. The project focuses on designing a supportive digital learning application for children with neurodevelopmental learning difficulties such as dyslexia, dyscalculia, dysgraphia, and working memory challenges.

The purpose of this questionnaire is to gather professional insights from therapists, educators, and specialists about current therapy activities, assessment methods, and effective learning strategies for children aged approximately 6–8 years.

The information collected will be used only for academic research and application development purposes. It will help us design a more effective, adaptive, and child-friendly learning application. All responses will be kept confidential and will only be used to improve the quality, usability, and effectiveness of the proposed mobile platform.

Questions

1. What types of activities or exercises are currently used in therapy for children with dyslexia, dyscalculia, dysgraphia, and working memory difficulties?
2. Are these activities usually paper-based, game-based, digital, or therapist-guided?
3. Which activities do you find most effective for children aged 6–8?
4. How do these activities change as the child improves or struggles?
5. Is there any recognized way to categorize these disorders into levels such as mild, moderate, or severe in clinical practice?
6. Do therapists assess children based on functional ability levels rather than biological markers?
7. How do you typically identify a child's current level — through standardized tests, observation, questionnaires, or activities?
8. If an app were to adjust its difficulty level based on a child's performance, what factors should it consider to avoid frustration or overstimulation?
9. What signs indicate a child should move to a higher or lower level?
10. Are there any activities or approaches that should be avoided for children with these disorders?
11. What are the signs that a child is overwhelmed rather than challenged?
12. What kind of progress information is most useful for parents?
13. How do therapists usually communicate improvement to parents?
14. From your professional experience, what is one feature you wish learning apps for neurodevelopmental disorders did better?

Risk Analysis

A risk matrix evaluating possible issues in the working memory support system, including the memory games, scoring engine, and backend services.

Risk Factor	Impact	Probability	Mitigation Strategy
Inaccurate memory performance measurement	High	Medium	Use multiple indicators such as accuracy, response time, and error rate to calculate performance instead of relying on a single metric.
Incorrect severity classification	High	Low	Test the weighted scoring algorithm with sample data and adjust threshold values based on expert feedback.
Children losing attention during games	Medium	High	Use short and engaging memory games with rewards, sounds, and animations to keep children motivated.
Data privacy and security risks	High	Low	Use secure login, encrypted connections, and protected database storage for children's data.
Internet connection issues	Medium	Medium	Allow gameplay to continue offline and synchronize data when the internet connection is available.

Plagiarism Report

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